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5 MILLION YEARS AFTER HUMANS GO EXTINCT



棲地

SUPRAGLACIAL ENVIRONMENT



棲地一：冰上環境

SUBGLACIAL LAKE



棲地二：冰下湖

冰上環境是指位於冰川或冰蓋最表層、直接暴露於大氣中的液態與半液態水體網絡（如冰川塵洞與冰上融水系），在強烈太陽輻射、高紫外線與極端凍融循環下，由光合自營生物驅動的動態開放生態系統。

冰上環境	
物理空間	數公里厚大陸冰原的最頂層，包含冰川塵洞、冰上融水湖河及表面雪層
溫度條件	大氣極寒環境-60 °C ~ 15 °C；塵洞與融水微棲地局部維持 -2°C ~ 2°C
化學物質	極度富氧、超低鹽度，富含大氣強風帶來的陸地礦物塵埃（鐵、矽、磷）與微量光化學過氧化物
能量來源	光合作用、大氣風降無機礦物質營養
生存危險	1 atm 大氣壓、冰雪雙重反射導致的高強度紫外線輻射暴曬、極夜驟降凍結、冰豎井融水急流沖刷

SUPRAGLACIAL ENVIRONMENT





SUBGLACIAL LAKE

冰下湖是指存在於冰川或冰蓋底部的液態水體，其形成與維持依賴於基底壓力融化、厚重冰層的熱絕緣作用以及地殼背景地熱流，在完全封閉、高壓與無光的極端條件下形成獨特的化能自營生態系統。

冰下湖）	
物理空間	位於 2 至 4 公里厚的大陸冰川正下方、基岩窪地中的深層封閉液態水體
溫度條件	-3.5 °C ~ 2.0 °C（冰點附近恆溫）
化學物質	極度缺氧、高濃度 CO ₂ 與 CH ₄ 、高濃度亞鐵離子 Fe ²⁺ 與冰川粉
能量來源	化能自營代謝
生存危險	350–400 atm 的超高水壓壓迫、極度營養匱乏與能量貧瘠、高濃度毒性氣體蓄積，以及冰川位移引發的洪水

冰下湖的生命存在 & 溫度和環境

Siegert MJ, Ellis-Evans JC, Tranter M, Mayer C, Petit JR, Salamatin A, Priscu JC. Physical, chemical and biological processes in Lake Vostok and other Antarctic subglacial lakes. Nature. 2001 Dec 6;414(6864):603-9. doi: 10.1038/414603a. PMID: 11740551.

Over 70 lakes have now been identified beneath the Antarctic ice sheet. Although water from none of the lakes has been sampled directly, analysis of lake ice frozen (accreted) to the underside of the ice sheet above Lake Vostok, the largest of these lakes, has allowed inferences to be made on lake water chemistry and has revealed small quantities of microbes. **These findings suggest that Lake Vostok is an extreme, yet viable, environment for life.** All subglacial lakes are subject to **high pressure (~350 atmospheres)**, **low temperatures (about -3°C)** and **permanent darkness**. Any microbes present must therefore **use chemical sources to power biological processes**. Importantly, **dissolved oxygen is available at least at the lake surface**, from equilibration with air hydrates released from melting basal glacier ice. Microbes found in Lake Vostok's accreted ice are relatively modern, but the probability of ancient lake or sediments leads to a possibility of a very old biota at the base of subglacial lakes.

#高壓：約 350 atm、低溫：約 -3°C、永久黑暗、湖中仍可能存在氧氣

In 1897, a group of Austrian artists led by Gustav Klimt broke from the conservative Academy of Fine Arts to form the Vienna Secession, a movement embracing Art Nouveau and Jugendstil (youth style)

化能自營生產量足以支撐異營代謝

Christner, B., Priscu, J., Achberger, A. et al. A microbial ecosystem beneath the West Antarctic ice sheet. Nature 512, 310–313 (2014). <https://doi.org/10.1038/nature13667>

Assuming that the thymidine and leucine values represent net incorporation, and that respiratory losses were 87% of net incorporation (i.e., average for Antarctic McMurdo Dry Valley lakes²⁴), the gross bacterial carbon demand (net productivity + respiration) would be 105 and 23 ng C L⁻¹ d⁻¹, respectively. If dark ¹⁴C-bicarbonate incorporation represents new organic carbon production via chemoautotrophy, the observed rates would meet between 31% and 143% of the heterotrophic carbon demand in the system. It should be noted that the effect of pressure (~ 8 MPa in SLW) was not tested and may influence the absolute rates of metabolism measured.

化能自營作用產生的新有機碳，理論上足以支撐相當大一部分的異營微生物代謝

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發現硝化相關微生物 + NH_4^+

Christner, B., Priscu, J., Achberger, A. et al. A microbial ecosystem beneath the West Antarctic ice sheet. Nature 512, 310–313 (2014). <https://doi.org/10.1038/nature13667>

Ammonium accounted for 73% of the dissolved inorganic nitrogen pool within the water column of SLW (Table 1). Given that mineral sources of ammonium are minor, the majority of the ammonium is likely from microbial mineralization. Soluble reactive phosphorus levels were similar to the total inorganic nitrogen pool (dissolved N:P molar ratio of 1.1), implying a biologically nitrogen deficient environment, relative to phosphorus. Unfortunately, sample limitations precluded measurement of dissolved organic N and P concentrations to assess their nutritional contribution. In addition to its nutritional role, ammonium is also an energy source for chemolithoautotrophic ammonium oxidizing bacteria and archaea. Evidence for complete nitrification in the aerobic SLW water column was supported by $\Delta^{17}\text{O}$ - NO_3 values (-0.1 ‰ to 0.2‰) that indicated microbial processes rather than atmospheric input was the dominant source for nitrate in the lake.

銨鹽佔了惠蘭斯冰下湖（SLW）水體中溶存無機氮（DIN）總量的 73%
由於礦物釋放的銨極少，因此推測大多數的銨鹽都是由微生物分解有機物（礦化作用）所產生的

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氮循環與碳循環連結

Christner, B., Priscu, J., Achberger, A. et al. A microbial ecosystem beneath the West Antarctic ice sheet. Nature 512, 310–313 (2014). <https://doi.org/10.1038/nature13667>

Our data show that SLW supports a metabolically active and phylogenetically diverse ecosystem that functions in the dark at subzero temperatures, confirming more than a decade of circumstantial evidence regarding the presence of life beneath Antarctica's ice sheet. Rate experiments revealed that chemoautotrophic primary production in SLW is adequate to support heterotrophic metabolism in the subglacial ecosystem. The abundance of taxa related to nitrifiers in concert with elevated ammonium and $\Delta^{17}\text{O}-\text{NO}_3$ values near 0 ‰ in the water column (Table 1) implies that nitrification may be a fundamental chemoautotrophic pathway of new organic carbon production in SLW. Similar conclusions regarding the ecological significance of nitrification have been drawn for the water column beneath the Ross Ice Shelf and in McMurdo Sound. Given the prevalence of subglacial water in Antarctica, our data from SLW lead us to contend that aquatic microbial ecosystems are common features of the subsurface environment that exists beneath the ~107 km² Antarctic Ice Sheet, which may have significant roles in stimulating Southern Ocean primary productivity.

- **SLW 中的微生物能利用無機物（特別是 NH_4^+ ）取得能量，透過硝化等化能自營作用固定無機碳、產生新的有機碳，進而支撐異營微生物的生態活動，形成沒有陽光也能運作的冰下微生物生態系統。**

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- **世界設定**

時間：人類消失後 500 萬年

地點：極地的冰下湖泊生態系，也可能加入火山／地熱環境

環境：厚冰覆蓋、缺乏陽光、低溫，冰下包含湖水、岩石、沉積物與冰下空腔等不同環境

生存挑戰：缺乏陽光導致能量來源有限與食物匱乏。在氮限制（N-LIMITATION）的環境下，初級生產者能否透過化學自營作用取得足夠能量，成為支撐後續食物網的基礎

- **為什麼選擇「500 萬年後」？**

靈感來自紀錄片《THE FUTURE IS WILD》。其中描寫五百萬年後的歐洲重新被冰川覆蓋，現在熟悉的許多動物已經消失，新的生物則演化出適應寒冷環境的生存方式。我喜歡這個時間尺度，五百萬年的時間足以讓環境與生物產生巨大改變，因此我選擇五百萬年後作為世界的時間背景。

- **為什麼選擇冰下世界？**

一開始我考慮過火山湖，但在查資料的過程中，我開始對冰下湖產生興趣。如果只是設定在冰層表面，我認為和現代極地環境的差異不夠大；相較之下，冰層下方是一個與地表隔離程度較高、較少受到人類直接活動影響的環境。

進一步了解後，我發現冰下湖的形成與維持本身就很特別。厚重冰層造成的壓力會降低冰的熔點，而來自地底的地熱也能提供熱量，使冰層底部即使位於極地仍可能維持液態水。

冰下環境也不只有水，還包含基岩、沉積物、冰下空腔與水體之間的物質交換。

如果這樣一個與地表不同、缺乏陽光卻仍存在液態水與物質循環的環境持續數百萬年，是否有可能逐漸形成更加複雜的生態系？因此目前我選擇冰下湖泊作為世界的主要地點。